



MFVC-DM: A Multi-Feature Vector Construction-based Method for Adversarial Example Detection in Automatic Modulation Classification

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Abstract

Deep neural network (DNN) based automatic modulation classification (AMC) has become increasingly important for wireless communication systems, yet these networks exhibit inherent vulnerabilities to adversarial perturbations that can severely degrade classification performance. Adversarial example detection offers an effective defense strategy by identifying malicious inputs before they reach the classifier, protecting AMC system security without architectural modifications. This paper proposes a multi-feature vector construction-based detection method (MFVC-DM) that integrates complementary statistical representations, including kernel density estimation, local intrinsic dimensionality, and K-means distance metrics. These features are extracted from hidden layer activations and cascaded into a unified high-dimensional feature space through dimension normalization and orthogonality. Experimental validation on the RML2016.10A dataset demonstrates that MFVC-DM consistently outperforms single-feature detection approaches, achieving up to 99.38% detection accuracy against gradient-based attacks and 93.43% against optimization-based perturbations, thereby providing an effective security layer for AMC systems.

Keywords Adversarial example detection · Adversarial attack · Deep neural network · Feature extraction

1 Introduction

With the proliferation of mobile devices, wireless communication systems have become essential for intelligent information processing in mobile multimedia environments. Automatic modulation classification (AMC) has emerged as a critical component in these systems, playing a pivotal role in spectrum monitoring, interference detection, and cognitive radio applications [1, 2]. The ability to automatically recognize and classify diverse modulation schemes has become indispensable for efficient spectrum management and signal intelligence in complex electromagnetic environments [3]. Deep neural networks (DNNs) have significantly advanced this field by enabling automatic learning of hierarchical representations directly from raw in-phase and quadrature (I/Q) samples without manual feature engineering [4]. Building upon decades of signal processing research, these data-driven approaches demonstrate remarkable classification accuracy across various signal-to-noise ratios (SNRs) and channel conditions [5].

Despite the significant advancements offered by DNNs, these models exhibit inherent vulnerabilities to adversarial attacks. Szegedy et al. [6] first demonstrated that

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imperceptible perturbations added to input samples could cause DNNs to misclassify with high confidence, revealing this inherent fragility of neural networks. This vulnerability extends to the signal domain, where AMC systems face substantial security challenges from adversarial attacks [7–9]. These attacks are increasingly feasible in practical wireless environments due to software-defined radio platforms that enable precise signal manipulation. As shown in Fig. 1, adversarial perturbations can reduce classification accuracy by orders of magnitude while requiring significantly less power than conventional jamming. Adversarial examples pose significant threats to wireless communication systems by compromising signal integrity through various attack vectors including direct classifier access and over-the-air transmission [10, 11].

To counter adversarial threats against DNN-based AMC systems, researchers have developed various defense strategies that encompass multiple complementary approaches. These defenses include model-enhancing techniques such as adversarial training and certified smoothing [12, 13], input preprocessing methods that apply transformations to neutralize perturbations, ensemble approaches that combine multiple models for improved robustness, and detection-based methods that identify adversarial examples before classification [14, 15]. Model enhancement and preprocessing techniques typically require modifications to the original classifier architecture or training process, which may impact computational efficiency and deployment flexibility [16]. In contrast, detection-based approaches operate as external filters without modifying the original model, offering an additional security layer that can be integrated with existing AMC systems while preserving their original functionality.

Adversarial example detection functions as a critical pre-classification filter that identifies malicious inputs without modifying the original model architecture [14]. In computer

vision, detection strategies broadly fall into statistical methods, activation space analysis, and auxiliary detector networks [17]. These approaches have been adapted to the radio frequency (RF) domain with several techniques, including peak-to-average power ratio-based statistical tests for waveform irregularities [18], multi-feature fusion combining features of the local intrinsic dimensionality (LID) and constellation diagram [19], and feature attribution methods that analyze model interpretation patterns [20]. Despite these advances, existing detection methods typically rely on single feature extraction techniques or limited feature combinations, failing to fully leverage the diverse statistical discrepancies between clean and adversarial examples across multiple domains. This underutilization leads to critical detection limitations, as single statistical measures may miss adversarial perturbations that manifest differently across feature spaces, resulting in reduced detection sensitivity for sophisticated attacks and increased false negative rates when adversarial examples exploit blind spots inherent to individual feature extractors. Consequently, current approaches perform inconsistently against different types of attacks, especially when attackers deliberately exploit the limitations of single-feature detection methods.

In order to solve the above problems, we proposed the multi-feature vector construction-based detection Method (MFVC-DM) for identifying adversarial examples in AMC. MFVC-DM utilizes a multi-statistical feature extraction mechanism encompassing kernel density (KD) estimation functions along with K-means (KM) distance metrics plus LID calculations to quantify the distinctive characteristics inherent in signal representations. The method subsequently employs a feature integration protocol involving dimensional normalization techniques and orthogonal vector concatenation operations to formulate a unified high-dimensional feature space. This feature integration methodology

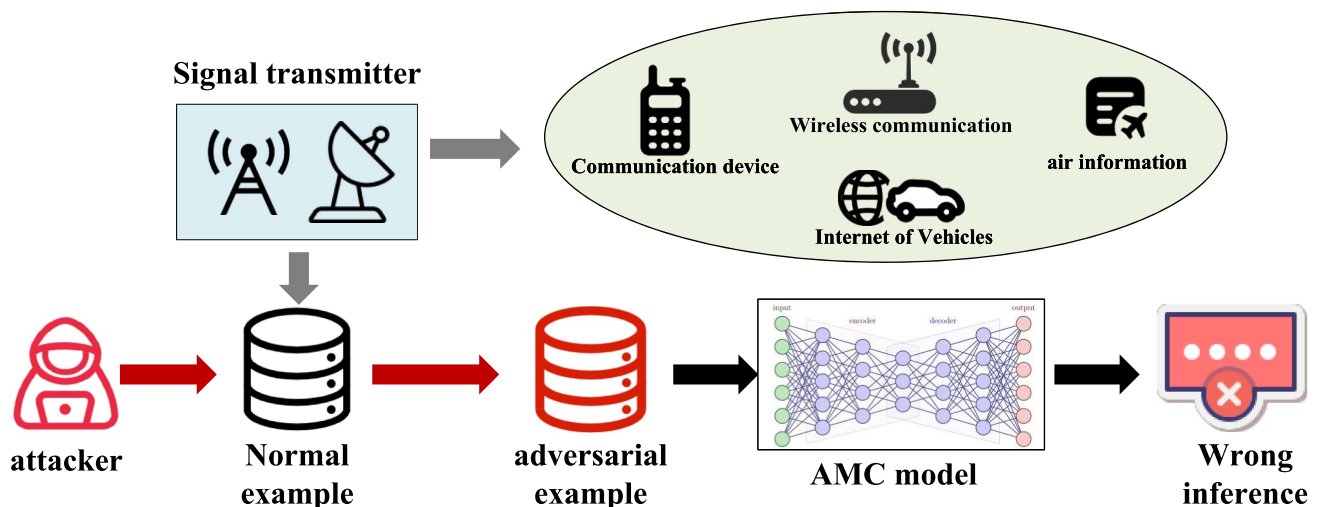


Fig. 1 Threat scenarios of adversarial examples in communication

synergistically leverages the discriminative characteristics from multiple statistical domains while maintaining their complementary information structures, thereby enhancing the system's capability to differentiate between clean and adversarial signal representations in the feature space. The primary contributions of this paper are as follows.

- An efficient multi-feature construction detection framework is proposed, which systematically integrates complementary statistical representations, to identify adversarial examples in AMC systems through exploitation of distinctive feature disparities between legitimate and malicious signal samples.
- A methodological feature fusion strategy is developed that combines KD estimation, KM distance, and LID metrics through dimensional normalization and orthogonal concatenation, thereby overcoming the inherent limitations of single-feature approaches while preserving the discriminative integrity of component representations in high-dimensional feature space.
- Extensive experimental validation is conducted on publicly available RF signal datasets, which demonstrates that MFVC-DM achieves higher detection rates than other methods while maintaining the lowest false positive rates (FPR) across multiple adversarial attack scenarios.

The rest of the paper is organized as follows. Section 2 surveys recent advances, Section 3 formulates the threat model of adversarial attack, Section 4 elaborates the proposed MFVC-DM mechanisms, Section 5 presents comprehensive experimental validation, and Section 6 concludes the paper with insights and directions for future research.

2 Related Work

2.1 Adversarial Attack

Adversarial attacks against DNNs have evolved through various mechanisms, including single-step methods that apply perturbations in one operation [21], iterative approaches that progressively refine perturbations [22, 23], and optimization techniques that minimize perturbation magnitude while ensuring misclassification [24]. In AMC applications, neural networks exhibit significant vulnerabilities to such attacks, with research showing that modulation classification models suffer substantial accuracy degradation under both white-box scenarios where attackers have full model access and black-box scenarios with limited system knowledge, requiring only small perturbations typically less than 10% of the original signal power [8, 9]. These vulnerabilities extend

beyond accuracy reduction to affect the overall reliability of DNN-based AMC systems, with experimental studies demonstrating that strategically crafted perturbations can cause exponential increases in transmission errors within communication systems that depend on accurate modulation recognition [11]. While defense mechanisms have been extensively studied in image processing, their adaptation to wireless signal processing introduces domain-specific challenges requiring specialized solutions [7], necessitating the development of robust defensive mechanisms specifically designed for wireless communication applications.

2.2 Adversarial Detection

Researchers have developed various methodologies for identifying adversarial examples across different domains. Pioneering research by Feinman et al. [25] explored statistical density measurements combined with uncertainty quantification techniques for distinguishing malicious inputs. Ma et al. [26] advanced the field by analyzing dimensional characteristics of adversarial manifolds. Within wireless communications, detection techniques have gained increasing attention. Kokalj-Filipovic et al. [18] implemented statistical analysis frameworks examining signal power distributions and classification confidence values. Davaslioglu and Sagduyu [27] focused on identifying compromised training samples through activation pattern analysis and statistical anomaly recognition. Xu et al. [19] contributed by integrating dimensional analysis with signal constellation properties for enhanced detection capabilities. However, most existing approaches rely on single statistical measures or limited feature combinations, which may fail to capture the full spectrum of anomalies that adversarial perturbations introduce across different feature spaces. This limitation motivates the exploration of multi-feature detection strategies that can leverage complementary statistical properties to improve detection accuracy.

3 Threat Model

Adversarial attacks targeting AMC architectures can be formally characterized through mathematical optimization with constraints. For a neural network classification system $f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y}$ parameterized by weights θ , the generation process for adversarial signal x_{adv} involves introducing carefully crafted disturbance δ to an authentic signal sample x , expressed as:

$$x_{adv} = x + \arg \max_{\|\delta\|_p \leq \epsilon} \mathcal{L}(f_{\theta}(x + \delta), y_{true}). \quad (1)$$

In this formulation, $\mathcal{L}$ represents the objective function maximizing classification error, y_{true} indicates the correct classification category, while the constraint $\|\delta\|_p \leq \epsilon$ establishes boundaries on disturbance magnitude measured through ℓ_p -norm metrics.

3.1 Adversarial Attack Methods

Common adversarial attacks can be divided into gradient-based attacks and optimization-based attacks based on their computational strategies and algorithmic complexity [28]. Gradient-based methods exploit neural network linearity by perturbing inputs along gradient directions. Fast gradient sign method (FGSM) [21] generates single-step adversarial examples through:

$$x_{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(f_\theta(x), y_{\text{true}})), \quad (2)$$

where ϵ controls the perturbation magnitude under the ℓ_∞ constraint. Basic iterative method (BIM) [22] extends this approach through iterative application of smaller perturbation steps with constraint clipping at each iteration, effectively refining the adversarial perturbation through multiple gradient updates. Projected gradient descent (PGD) [23] further enhances attack robustness by incorporating random initialization within the perturbation budget and employing projection operations to maintain strict adherence to the ℓ_p -norm constraint at each iteration.

Optimization-based attacks formulate adversarial generation as constrained minimization problems seeking minimal perturbations that induce misclassification. The Carlini & Wagner (CW) [24] attack utilizes a dual-objective optimization framework that jointly minimizes perturbation magnitude and maximizes confidence in targeted misclassification through operations on pre-softmax logit representations. DeepFool [29] employs geometric optimization by iteratively computing orthogonal projections to linearized decision boundaries, finding the shortest perturbation path that

crosses classification boundaries. Both methods generate adversarial examples through iterative optimization procedures that prioritize perturbation minimality while ensuring successful misclassification.

3.2 Detection of Adversarial Examples

Adversarial example detection constitutes a binary classification paradigm that differentiates between legitimate signals and maliciously perturbed inputs within wireless communication systems. The detection framework can be formalized as:

$$\mathcal{D}(x) = \begin{cases} 0, & \text{if } x \in \mathcal{X}_{\text{clean}}, \\ 1, & \text{if } x \in \mathcal{X}_{\text{adversarial}}, \end{cases} \quad (3)$$

where $\mathcal{D} : \mathcal{X} \rightarrow \{0, 1\}$ represents the detector function mapping input signal space $\mathcal{X}$ to binary classification outcomes. The detector operates by extracting discriminative statistical representations that capture perturbation-induced anomalies across multiple signal dimensions. These extracted features characterize distributional shifts between pristine and adversarial RF signals, enabling effective differentiation prior to modulation classification.

4 Methodology

4.1 The Overall MFVC-DM Framework

The overall framework of proposed MFVC-DM is illustrated in Fig. 2. First, the feature extraction module processes input signals through a DNN to generate multi-domain statistical representations via parallel computation of LID, KD estimation, and KM distance metrics, each quantifying distinctive distributional characteristics between legitimate and adversarial samples at different granularities. These extracted feature sets subsequently undergo standardization

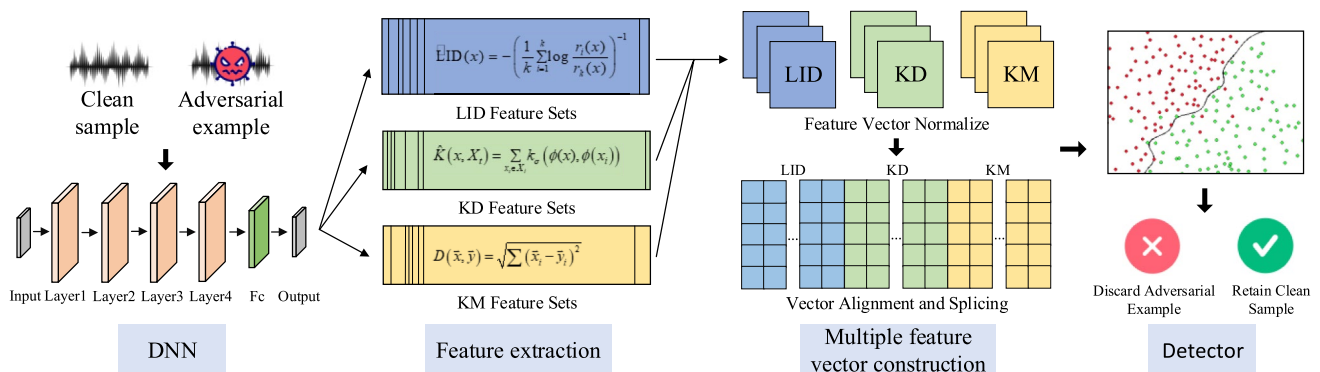


Fig. 2 The proposed MFVC-DM framework

preprocessing to ensure dimensional consistency, followed by orthogonal alignment and concatenation operations that construct a unified high-dimensional composite representation preserving complementary discriminative information from multiple statistical perspectives. This integrated feature vector, encapsulating collective statistical anomalies induced by adversarial perturbations while maintaining structural integrity across feature subspaces, is finally processed by a binary classifier that enables the system to filter malicious examples before they propagate through the modulation recognition pipeline, effectively providing an additional security layer without modifying the original classification architecture.

4.2 Feature Extraction

4.2.1 Kernel Density Estimation Features

KD estimation facilitates the quantification of distributional anomalies within the feature space of adversarial examples by examining their proximity to manifolds of legitimate class distributions. We define $\phi(x)$ as the mapping function that extracts the activation vector from the terminal hidden layer of the DNN for input sample x . For each sample x , we compute density estimates relative to each class manifold c_x , enabling the detection of adversarial perturbations through their deviation from natural sample distributions. Leveraging the capacity of DNNs to generate increasingly linearized and expanded manifolds in higher-dimensional spaces, we implement density estimation within the feature representation at the terminal hidden layer. The KD estimation formulation is expressed as:

$$\hat{f}(x) = \frac{1}{|X_t|} \sum_{x_i \in X_t} k(x_i, x), \quad (4)$$

where X_t represents the set of samples predicted as class t , and $k(\cdot)$ denotes the kernel function, typically selected as a Gaussian kernel with bandwidth σ :

$$k_\sigma(x, y) \sim \exp\left(-\frac{\|x - y\|^2}{\sigma^2}\right), \quad (5)$$

Applying density estimation within the terminal hidden layer's representation space, the density estimate for a sample x relative to predicted class t is defined as:

$$\hat{K}(x, X_t) = \sum_{x_i \in X_t} k_\sigma(\phi(x), \phi(x_i)), \quad (6)$$

where $\phi(x)$ represents the activation vector from the terminal hidden layer, and σ constitutes the bandwidth parameter.

This approach exhibits high detection efficacy for adversarial examples residing distant from their targeted class manifolds, though its effectiveness diminishes for adversarial examples positioned in proximity to targeted class manifolds.

4.2.2 Local Intrinsic Dimensionality Features

The LID metric quantifies topological properties surrounding datapoints within high-dimensional embeddings. Consider a sample vector $x \in X$ with proximity relationships to neighboring points represented by random variable $R > 0$. When the cumulative probability function $F(r)$ exhibits both continuity and differentiability at any radius $r > 0$, the dimensional measure at x for neighborhood radius r can be expressed through the expansion rate formulation:

$$\text{LID}_F(r) \triangleq \lim_{\epsilon \rightarrow 0} \frac{\ln(F((1 + \epsilon) \cdot r)/F(r))}{\ln(1 + \epsilon)} = \frac{r \cdot F'(r)}{F(r)}. \quad (7)$$

Statistical foundations for practical estimation derive from asymptotic distribution analysis. Specifically, nearest-neighbor distance sequences (selecting k minimal distances) correspond to minimal extrema in probability theory, representing rare events in the distribution's lower boundary region. When appropriate smoothness conditions are satisfied, these minimal distance values asymptotically conform to transformed generalized Pareto distribution patterns. By leveraging this theoretical correspondence, computational implementation employs maximum likelihood estimation techniques to balance analytical precision with processing efficiency. For any observation x sampled from distribution P , the practical estimator formulation for dimensional complexity measurement is constructed as:

$$\text{LID}(x) = -\left(\frac{1}{k} \sum_{i=1}^k \log \frac{r_i(x)}{r_k(x)}\right)^{-1}, \quad (8)$$

where $r_i(x)$ denotes the distance between x and its i -th nearest neighbor, and $r_k(x)$ represents the maximum of these k distances. This estimator captures the dimensional complexity of the local region surrounding each sample, providing a powerful discriminative feature for detecting adversarial perturbations in the RF signal space.

4.2.3 K-means Distance Features

KM distance methodology constitutes a clustering analysis approach where k sample points are randomly selected as initial cluster centroids, represented

as $m^{(0)} = (m_1^{(0)}, \dots, m_i^{(0)}, \dots, m_k^{(0)})$. For fixed cluster centroids $m^{(t)} = (m_1^{(t)}, \dots, m_i^{(t)}, \dots, m_k^{(t)})$, distances between each sample and centroids are computed, with samples subsequently assigned to the nearest cluster centroid, forming new clustering results $C^{(t)}$. The average of samples within each cluster is calculated as the new cluster centroid $m^{(t+1)} = (m_1^{(t+1)}, \dots, m_i^{(t+1)}, \dots, m_k^{(t+1)})$, iteratively refined until convergence or satisfaction of termination criteria. The objective function is defined as:

$$E = \sum_{i=1}^k \sum_{p \in C_i^t} \|p - m_i\|^2, \quad (9)$$

where E represents the sum of squared distances between samples and their assigned cluster centroids, and p denotes individual data points within cluster C_i^t . The distance metric predominantly employed in KM analysis is the Euclidean distance. Given two n -dimensional vectors x and y , their Euclidean distance is calculated as:

$$D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}, \quad (10)$$

For detection purposes, we extract KM distance features by computing the minimum distance from each sample to all cluster centroids, defined as $F_{KM}(x) = \min_{i=1, \dots, k} \|\phi(x) - m_i\|$, where $\phi(x)$ represents the feature representation from the terminal hidden layer. This distance metric serves as a fundamental feature for adversarial example detection by quantifying distributional shifts in the feature space induced by adversarial perturbations. Adversarial examples typically exhibit larger minimum distances to cluster centroids compared to clean samples, enabling effective detection through this geometric anomaly.

4.3 Multiple Feature Vector Construction

The proposed detection methodology first generates adversarial examples x_{adv} through five distinct attack techniques applied to original samples x . Subsequently, three complementary statistical features, namely KD, LID, and KM distance, are extracted from both original and adversarial examples. These feature sets are integrated to train a binary classifier that functions as the detection mechanism.

To overcome the inherent limitations of single-feature approaches in adversarial example detection, we implement

a feature fusion strategy that constructs a comprehensive high-dimensional composite representation. This fusion methodology concatenates feature vectors along their dimensional axes, formulated as:

$$\vec{v}_{MFVC} = [\vec{v}_{KD}; \vec{v}_{KM}; \vec{v}_{LID}], \quad (11)$$

where $\vec{v}_{KD}$ denotes the KD estimation feature vector, $\vec{v}_{KM}$ represents the KM distance feature vector, and $\vec{v}_{LID}$ corresponds to the LID feature vector. Prior to concatenation, each feature set undergoes standardization via Z-Score normalization to ensure dimensional consistency:

$$x^* = \frac{x - \text{mean}(x)}{\text{std}(x)}, \quad (12)$$

where $\text{mean}(x)$ and $\text{std}(x)$ represent the mean and standard deviation of variable x across the sample distribution.

The dimensional alignment process verifies consistent sample ordering across feature sets to maintain coherence and completeness in the resulting high-dimensional representation. The integrated feature vector serves as input to the detection classifier, which employs binary cross-entropy loss:

$$L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})], \quad (13)$$

where $\hat{y}$ denotes the predicted probability and y represents the ground truth label. This approach effectively leverages complementary statistical information, and the overall process of MFVC-DM is shown in Algorithm 1.

Algorithm 1 MFVC-DM: multi-feature vector construction-based detection method.

Require: Original samples $\mathcal{X}_{orig}$, DNN classifier f_θ , Parameters k, σ
Ensure: Adversarial example detector $\mathcal{D}$

- 1: Generate adversarial examples $\mathcal{X}_{adv} = \bigcup_{\mathcal{A} \in \{\text{FGSM, BIM, PGD, CW, DeepFool}\}} \mathcal{A}(\mathcal{X}_{orig}, f_\theta)$
- 2: Construct dataset $\mathcal{D}_{train} \leftarrow \{(\mathbf{x}, 0) | \mathbf{x} \in \mathcal{X}_{orig}\} \cup \{(\mathbf{x}_{adv}, 1) | \mathbf{x}_{adv} \in \mathcal{X}_{adv}\}$
- 3: **for** each $\mathbf{x} \in \mathcal{D}_{train}$ **do**
- 4: Extract activation vector $\phi(\mathbf{x})$ from terminal hidden layer of f_θ
- 5: $\vec{v}_{KD}(\mathbf{x}) \leftarrow [F_{KD}(\phi(\mathbf{x}), X_1), \dots, F_{KD}(\phi(\mathbf{x}), X_C)]$ ▷ Kernel density
- 6: $\vec{v}_{LID}(\mathbf{x}) \leftarrow F_{LID}(\phi(\mathbf{x}), k)$ ▷ Local intrinsic dimensionality
- 7: $\vec{v}_{KM}(\mathbf{x}) \leftarrow F_{KM}(\phi(\mathbf{x}), k)$ ▷ K-means distance
- 8: **end for**
- 9: Standardize features: $\vec{v}_{\mathcal{F}}^* \leftarrow \frac{\vec{v}_{\mathcal{F}} - \mu(\vec{v}_{\mathcal{F}})}{\sigma(\vec{v}_{\mathcal{F}})}$ for $\mathcal{F} \in \{\text{KD, LID, KM}\}$
- 10: $\vec{v}_{MFVC}(\mathbf{x}) \leftarrow [\vec{v}_{KD}^*(\mathbf{x}); \vec{v}_{KM}^*(\mathbf{x}); \vec{v}_{LID}^*(\mathbf{x})]$
- 11: Train binary classifier $\mathcal{D}$ with $\vec{v}_{MFVC}$ using loss function $\mathcal{L}(y, \hat{y}) = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$
- 12: **return** $\mathcal{D}$

5 Experimental Settings and Results Analysis

5.1 Datasets

This study employs the RML2016.10A dataset [30] for evaluating the proposed detection methodology. The dataset consists of 220,000 signal samples across 11 modulation schemes including both digital and analog variants, generated at SNR levels from -20dB to 18dB in 2dB increments. Each sample comprises 128 complex-valued I/Q samples representing baseband signals after appropriate preprocessing. Our experiments utilize exclusively the 0dB SNR subset, as it represents the threshold condition where signal power equals noise power. The dataset undergoes an 8:2 partition ratio into training and testing sets to ensure statistical validity while maintaining sufficient samples for neural network training.

5.2 Model

The AMC task is implemented using a convolutional neural networks (CNN) architecture, which serves as the foundation for both our baseline performance evaluation and adversarial example generation, which is shown in Fig. 3. The network accepts complex-valued I/Q samples of dimension 2×128 as input, representing in-phase and quadrature components of the baseband signal. The architecture comprises four sequential convolutional blocks with increasing filter counts of 32, 64, 128, and 256 respectively, maintaining consistent temporal resolution across layers. Each convolutional block incorporates batch normalization for training stability, followed by ReLU activation to introduce non-linearity into the model. The feature extraction hierarchy culminates in a flattened representation that feeds into a series of fully connected layers with dimensions 4096 and 512, progressively distilling signal characteristics into discriminative features. The output layer consists of 11 neurons corresponding to the distinct modulation schemes present

in the dataset, with softmax activation yielding probability distributions across possible classes.

5.3 Metrics

Performance evaluation of adversarial example detection methodologies requires rigorous quantitative assessment through multiple complementary metrics. This study employs a comprehensive evaluation framework comprising accuracy, precision, recall, FPR, and area under the curve (AUC).

Receiver operating characteristic (ROC) curve and its corresponding AUC serve as critical evaluation metrics in adversarial detection frameworks. ROC characterizes the discriminative capability of the detector by plotting true positive rate against FPR across multiple decision thresholds. AUC provides a scalar quantification of detector performance through the mathematical formulation:

$$\text{AUC} = \frac{\sum_{i=1}^{N_+} \sum_{j=1}^{N_-} I(f(x_i^+) > f(x_j^-))}{N_+ \cdot N_-}, \quad (14)$$

where N_+ and N_- denote adversarial and benign sample quantities respectively, $f(x)$ represents the classifier's prediction score, and $I(\cdot)$ is the indicator function. This threshold-invariant metric effectively quantifies detection performance regardless of class distribution imbalances, establishing it as a fundamental criterion for evaluating adversarial robustness in modulation classification systems.

5.4 Configuration

All experiments were conducted on a workstation equipped with an 8-core CPU operating at 2.50GHz, 64GB DDR4 dual-channel memory, and a graphics processing unit with 24GB GDDR6X memory. The proposed MFVC-DM framework and baseline methods were implemented using PyTorch 1.13 deep learning framework on a 64-bit operating system.

Fig. 3 The proposed CNN architecture

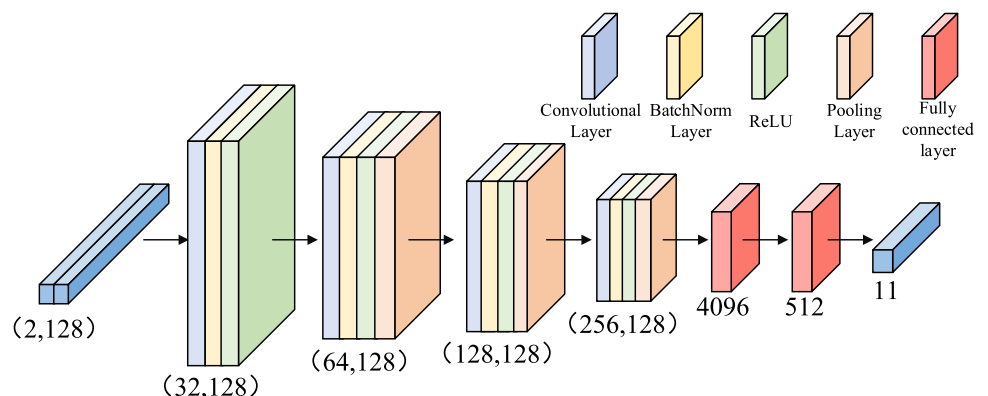


Table 1 Detection performance of MFVC-DM under various attacks

Attack	ϵ	Accuracy(%)	Precision(%)	Recall(%)	FPR(%)
FGSM	0.1	83.99	88.40	78.23	9.26
	0.2	89.72	92.55	86.39	6.95
	0.3	94.67	96.20	93.01	3.67
	0.4	98.38	98.75	97.99	1.24
	0.5	98.91	99.46	98.35	0.53
PGD	0.1	91.22	94.23	87.82	5.38
	0.2	96.69	98.59	94.74	1.36
	0.3	96.17	99.31	92.99	0.65
	0.4	99.38	99.70	99.05	0.30
	0.5	99.29	99.94	98.64	0.06
BIM	0.1	91.92	94.16	89.38	5.55
	0.2	96.27	99.18	93.32	0.77
	0.3	96.82	99.56	94.05	0.41
	0.4	98.12	99.76	96.47	0.24
	0.5	98.23	99.70	96.76	0.29
CW	—	70.51	75.97	59.99	8.97
DeepFool	—	93.43	97.29	89.34	2.49

5.5 Detection Capability Analysis under Various Attack

Table 1 presents the quantitative evaluation of MFVC-DM detection performance across diverse adversarial attack methodologies. For gradient-based attacks, detection efficacy increases monotonically with perturbation strength ϵ , as evidenced by FGSM accuracy improving from 83.99% at perturbation strength $\epsilon = 0.1$ to 98.91% when the perturbation strength increases to $\epsilon = 0.5$. This occurs because larger perturbations create more pronounced statistical anomalies that are easier to detect through our multi-feature approach. PGD attacks yield the highest detection rates with accuracy reaching 99.38% and precision at 99.94%, while maintaining a minimal FPR of 0.06% at maximum perturbation levels. BIM demonstrates intermediate performance characteristics with comparable detection trends. The superior performance against iterative attacks like PGD and BIM compared to single-step FGSM stems from their iterative refinement process, which tends to push adversarial examples further from the natural data distribution, making them more detectable.

Optimization-based attacks exhibit distinct detection characteristics, with CW achieving 70.51% accuracy and DeepFool attaining 93.43%, reflecting their fundamental algorithmic differences. The significantly lower performance against CW attacks results from its sophisticated optimization that explicitly minimizes perturbation magnitude while ensuring misclassification, producing adversarial examples that remain closer to legitimate signal distributions and thus harder to distinguish. In contrast, DeepFool's geometric approach of finding the shortest path to decision boundaries still introduces sufficient distributional changes

to be captured by our detection features. Across all attack vectors, MFVC-DM maintains FPR below 9.26%, demonstrating the robustness of the multi-feature fusion strategy. This performance superiority stems from the complementary nature of KD, KM, and LID features, which collectively capture diverse statistical anomalies that individually might remain undetected. The experimental results validate the framework's effectiveness in discriminating between legitimate and adversarial signals across both gradient-based and optimization-based attack paradigms.

5.6 Comparative Assessment of Detection Capabilities

This section evaluates MFVC-DM against established single-feature detection approaches BU [25], LID [18], KD [25], and KM [31] under standardized adversarial conditions.

5.6.1 Quantitative Assessment of Basic Detection Capabilities

The comparative framework employs gradient-based attacks with $\epsilon = 0.5$ and optimization-based attacks to quantify detection performance across diverse adversarial generation mechanisms in this section. Table 2 illustrates the quantitative metrics of MFVC-DM against baseline feature extraction methodologies across multiple adversarial attack vectors. For gradient-based perturbations, MFVC-DM achieves accuracy values of 98.91% for FGSM, 99.29% for PGD, and 98.23% for BIM, exceeding single-feature detectors by margins of 2.70%, 1.96%, and 1.65% respectively. Optimization-based attacks exhibit reduced detection accuracy across all methodologies, with MFVC-DM attaining 70.51% for CW and 93.43% for DeepFool. The significantly lower performance against CW attacks stems from its L2-norm optimization objective that produces smoother perturbations more closely aligned with natural data distributions, making statistical anomalies less pronounced in our feature space. In contrast, gradient-based attacks using Lp-norm constraints tend to create more detectable distributional shifts, while DeepFool's geometric approach, despite seeking minimal perturbations, still introduces sufficient changes in local manifold geometry to be captured by our multi-feature framework.

The precision-recall distribution demonstrates method-specific detection characteristics. MFVC-DM precision ranges from 99.46% to 99.94% for gradient-based attacks while maintaining recall values between 96.76% and 98.64%. KD features display heightened sensitivity with maximum recall values of 98.22% for PGD, 96.35% for BIM, 89.88% for CW, and 96.46% for DeepFool. This

Table 2 Comparative performance analysis of detection methods against various adversarial attacks

Attack	Metric	BU	LID	KD	KM	MFVC-DM
FGSM	Accuracy (%)	52.33	88.35	93.88	96.21	98.91
	Precision (%)	52.01	86.79	92.53	97.21	99.46
	Recall (%)	60.11	90.46	95.47	95.15	98.35
	FPR (%)	55.45	13.77	7.71	2.73	0.53
PGD	Accuracy (%)	64.90	89.22	97.33	91.31	99.29
	Precision (%)	69.69	96.62	96.51	99.88	99.94
	Recall (%)	52.73	81.28	98.22	82.73	98.64
	FPR (%)	22.93	2.84	3.56	0.10	0.06
BIM	Accuracy (%)	73.08	96.58	94.42	80.51	98.23
	Precision (%)	81.71	96.59	95.87	98.83	99.70
	Recall (%)	59.47	91.86	96.35	61.11	96.76
	FPR (%)	13.31	3.24	4.19	0.40	0.29
CW	Accuracy (%)	48.34	63.33	68.52	48.74	70.51
	Precision (%)	47.29	65.27	62.98	49.02	75.97
	Recall (%)	29.06	56.99	89.88	63.33	85.99
	FPR (%)	32.38	30.32	52.84	65.86	18.97
DeepFool	Accuracy (%)	51.86	79.77	88.81	90.61	93.43
	Precision (%)	51.56	81.12	83.66	92.58	97.29
	Recall (%)	61.59	77.60	96.46	88.28	94.34
	FPR (%)	57.88	18.06	18.84	7.07	2.49

recall advantage occurs at precision cost, particularly evident in the 62.98% precision against CW attacks. The feature fusion approach balances this precision-recall trade-off through multi-perspective adversarial characterization. And FPR analysis indicates classification reliability under normal signal conditions. MFVC-DM demonstrates FPR values of 0.53% for FGSM, 0.06% for PGD, and 0.29% for BIM, which represents the lowest error rate among all the methods.

5.6.2 Comparison of Detection Performance Based on ROC

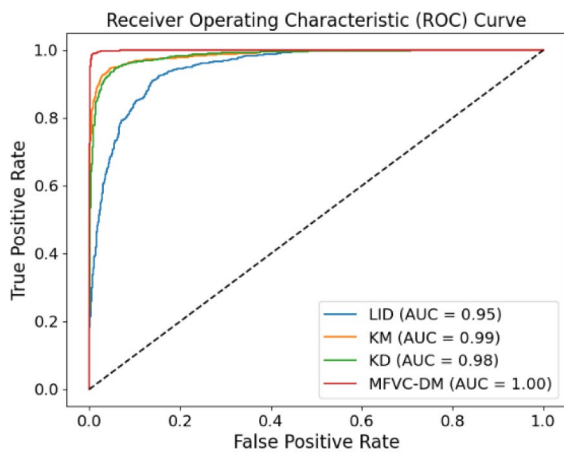
The ROC analysis quantifies detection efficacy across varying discrimination thresholds, revealing distinct performance gradients among feature extraction methodologies, which is shown in Fig. 4. For gradient-based perturbations, MFVC-DM achieves optimal AUC values of 1.00 across FGSM, PGD, and BIM vector spaces, exceeding the performance boundaries of singular feature extractors, which register AUC metrics ranging from 0.95 to 1.00. The multi-feature architecture demonstrates accelerated ROC trajectory convergence toward maximum true positive rates within minimal false positive operational domains, indicating enhanced manifold separation between clean and adversarial signal distributions. Under optimization-based attack conditions, MFVC-DM maintains consistent detection advantages with AUC values of 0.79 and 0.98 for CW and DeepFool respectively, surpassing LID, KM, and KD methodologies by AUC margins spanning 0.05 to 0.12 despite the intrinsic detection challenges posed by minimized perturbation vectors. The quantitative metrics indicate that the

multi-feature fusion architecture of MFVC-DM achieves consistent AUC improvements across both gradient-derived and optimization-based adversarial examples. This performance can be attributed to the method's parallel extraction of heterogeneous feature vectors that capture LID, KD, and manifold distance characteristics simultaneously, enabling the detection framework to identify complementary statistical anomalies that manifest differently across feature spaces but collectively signal adversarial perturbation presence.

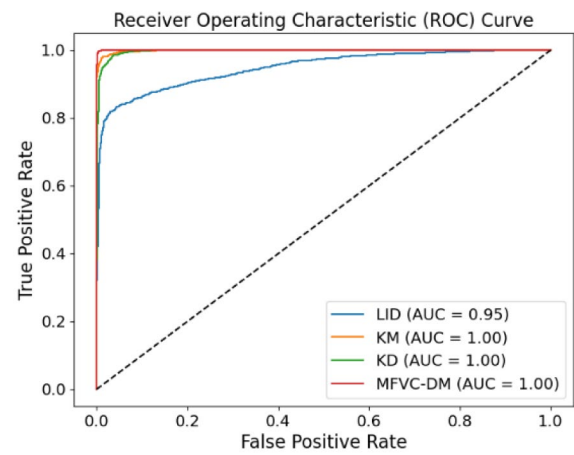
5.7 Visualization of Sample Distribution Characteristics

This section examines the statistical separation capabilities of the proposed multi-feature framework through rigorous distribution analysis, and Fig. 5 illustrates the characteristic distributions of four detection features. The adversarial examples were generated using the PGD attack algorithm with perturbation magnitude $\varepsilon = 0.2$. This empirical analysis quantifies the discriminative capacity of individual detection features.

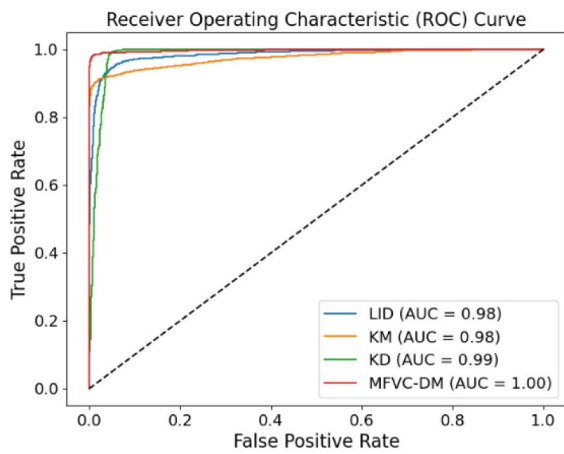
Figure 5(a) reveals that BU exhibits substantial overlap between sample types with near-identical median values 0.05-0.10, rendering it ineffective for adversarial detection despite its theoretical foundation in confidence estimation. In contrast, Fig. 5(b) demonstrates that KD scores achieve significant separation with median values of 0.92 for clean versus 0.98 for adversarial examples, confirming that adversarial examples typically reside in low-density regions of the feature manifold. Figure 5(c) illustrates the most pronounced separation through LID, where clean samples



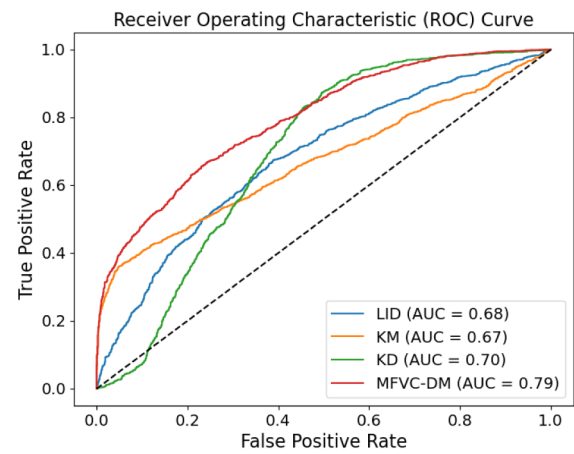
(a) FGSM



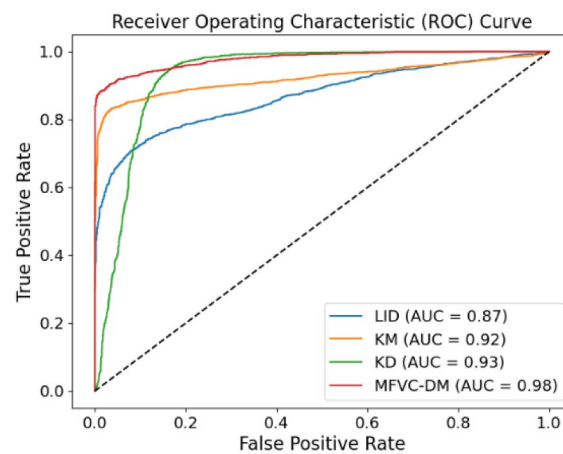
(b) PGD



(c) BIM



(d) CW



(e) DeepFool

Fig. 4 ROC curves of various detection methods under different attacks

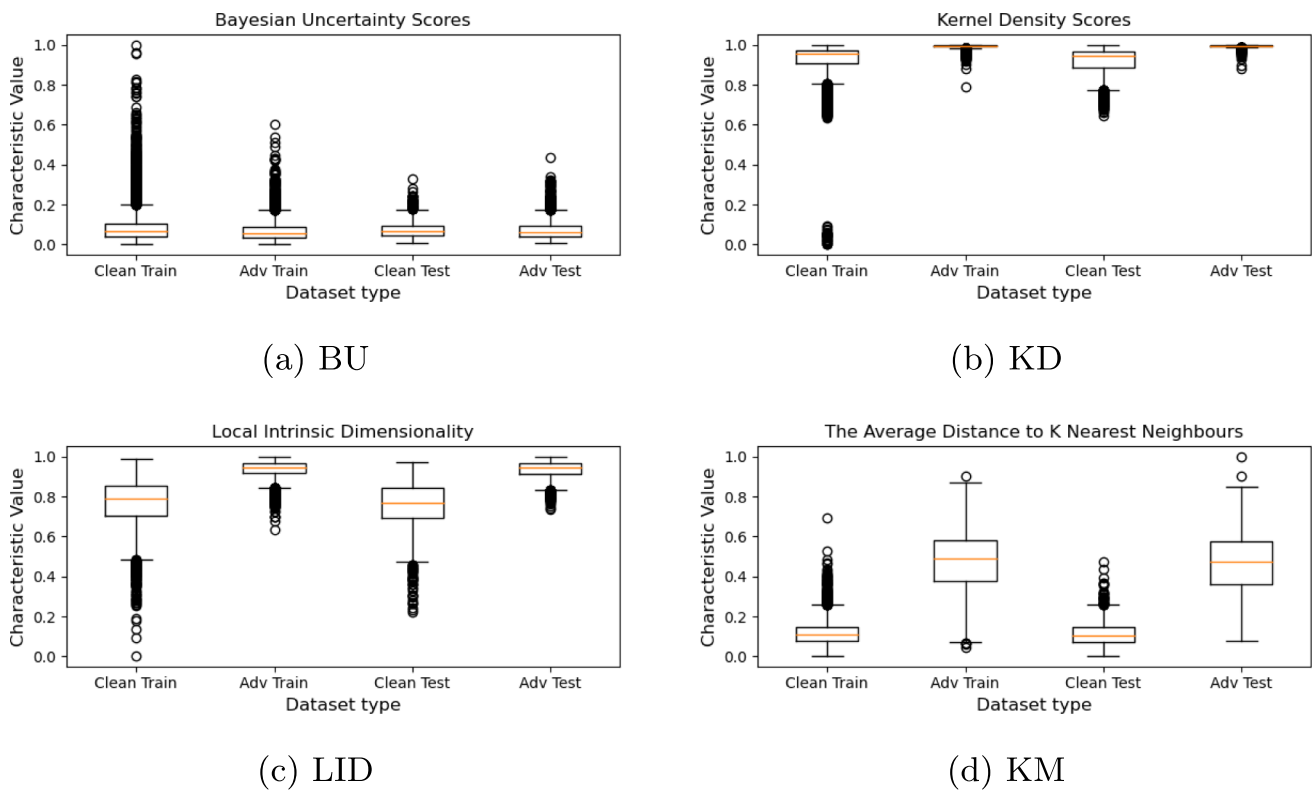


Fig. 5 Visualize the distribution characteristics of the four features on clean examples and adversarial examples based on box plots

concentrate within 0.68–0.83 while adversarial examples occupy 0.85–0.98, validating that adversarial perturbations cause abnormal expansion in local manifold dimensionality. Figure 5(d) shows KM further reinforcing detection capability with median values of 0.52 for adversarial examples versus 0.12 for clean examples, quantifying the spatial displacement induced by adversarial perturbations. This complementary coverage explains the ideal performance of the combined methodology over single-feature approaches.

6 Conclusion

This paper presents MFVC-DM, a multi-feature vector construction-based detection framework for identifying adversarial examples in AMC systems. The methodology leverages complementary statistical representations, including KD estimation, LID, and KM distance metrics, to characterize distributional anomalies induced by adversarial perturbations in the feature space. Through feature fusion via standardization and orthogonal concatenation, MFVC-DM constructs a unified high-dimensional representation that captures multifaceted adversarial characteristics while preserving feature complementarity. Quantitative analysis reveals that MFVC-DM consistently outperforms single-feature detection approaches and achieves substantial FPR

reduction. Future research directions include extending the multi-feature detection framework to other wireless communication tasks such as channel estimation and beamforming, investigating adaptive detection mechanisms that can evolve with emerging attack strategies, and exploring the integration of federated learning approaches for distributed adversarial detection across multiple communication nodes without compromising data privacy.

Author Contributions Z.B. and J.F. conceived the research idea and designed the methodology, Z.B. and C.Y. implemented the experiments and analyzed the data, J.L., P.Z., and C.L. contributed to algorithm development and validation, Z.B. and C.Y. wrote the main manuscript text, and all authors reviewed and approved the final manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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